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manchodu@Srini:~$ sudo ip netns add host1
[sudo] password for manchodu:
Cannot create namespace file "/run/netns/host1": File exists
manchodu@Srini:~$
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manchodu@Srini:~$ sudo ip netns add host1
[sudo] password for manchodu:
Cannot create namespace file "/run/netns/host1": File exists
manchodu@Srini:~$ sudo ip netns add router
Cannot create namespace file "/run/netns/router": File exists
manchodu@Srini:~$ sudo ip netns add host2
Cannot create namespace file "/run/netns/host2": File exists
manchodu@Srini:~$
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```
manchodu@Srini:~$ sudo ip netns add host1
[sudo] password for manchodu:
Cannot create namespace file "/run/netns/host1": File exists
manchodu@Srini:~$ sudo ip netns add router
Cannot create namespace file "/run/netns/router": File exists
manchodu@Srini:~$ sudo ip netns add host2
Cannot create namespace file "/run/netns/host2": File exists
manchodu@Srini:~$ sudo ip link add veth-h1 type veth peer name veth-r1
manchodu@Srini:~$ sudo ip link add veth-r2 type veth peer name veth-h2
manchodu@Srini:~$ sudo ip link set veth-h1 netns host1
manchodu@Srini:~$ sudo ip link set veth-r1 netns router
manchodu@Srini:~$ sudo ip link set veth-r2 netns router
manchodu@Srini:~$
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manchodu@Srini:~$ sudo ip netns exec host1 ip link set veth-h1 up
[sudo] password for manchodu:
manchodu@Srini:~$ sudo ip netns exec host1 ip link set veth-h1 up
manchodu@Srini:~$ sudo ip netns exec host1 ip link set lo up
manchodu@Srini:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
manchodu@Srini:~$ sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
manchodu@Srini:~$ sudo ip netns exec router ip link set veth-r1 up
manchodu@Srini:~$ sudo ip netns exec router ip link set veth-r2 up
manchodu@Srini:~$ sudo ip netns exec router ip link set lo up
manchodu@Srini:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
manchodu@Srini:~$ sudo ip netns exec host2 ip link set veth-h2 up
manchodu@Srini:~$ sudo ip netns exec host2 ip link set lo up
manchodu@Srini:~$
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manchodu@Srini:~$ sudo ip netns exec host2 ip link set lo up
manchodu@Srini:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
manchodu@Srini:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
manchodu@Srini:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
manchodu@Srini:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.083 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.117 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.058 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 1998ms
rtt min/avg/max/mdev = 0.058/0.086/0.117/0.024 ms
manchodu@Srini:~$
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manchodu@Srini:~$ sudo ip netns exec host2 ip link set veth-h2 up
manchodu@Srini:~$ sudo ip netns exec host2 ip link set lo up
manchodu@Srini:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
manchodu@Srini:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
manchodu@Srini:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
manchodu@Srini:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.083 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.117 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.058 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 1998ms
rtt min/avg/max/mdev = 0.058/0.086/0.117/0.024 ms
manchodu@Srini:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
manchodu@Srini:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 13 limit 1000 delay 20ms loss 1%
manchodu@Srini:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
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